



Career Trajectory Optimizer

A Deep Agents multi-agent career coach for Indian tech professionals

An orchestrator plans, detects intent, and delegates to eight specialist subagents — profile parsing, live market research, salary benchmarking, market tracking, risk scoring, goal-based recommendation, roadmap planning, and critique — sharing state through a composite filesystem + store backend, with a human-in-the-loop gate before any simulated calendar sync.

Python 3.12+

LangChain/LangGraph

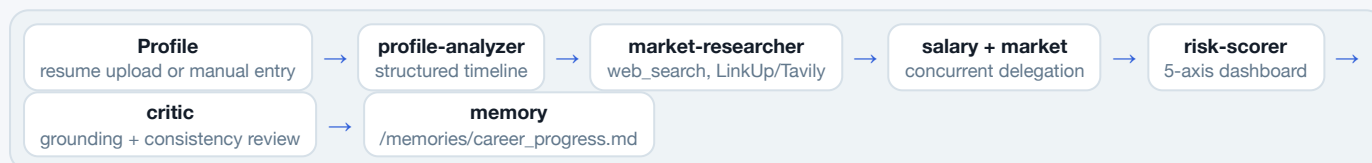
Deep Agents

OpenAI

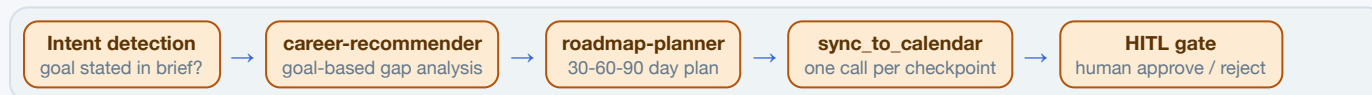
Streamlit

What it is. A Deep Agents application: one orchestrator LLM plans with `write_todos`, detects whether the user stated a career goal, and delegates isolated, stateless subagent runs via the `task` tool. Findings pass between subagents through a shared virtual filesystem (`CompositeBackend` : disk for `/profile` /`research` /`analysis` /`roadmap`, a `Store` for `/memories`). A checkpointer enables a pause-and-resume human approval gate before the (simulated) calendar-sync action.

SHARED PIPELINE (EVERY RUN)



CAREER ASPIRATION MODE ONLY



Subagents & division of labour

- **Eight specialists, one orchestrator.** profile-analyzer, market-researcher, salary-intelligence, market-tracker, risk-scorer, career-recommender, roadmap-planner, critic — each with its own prompt, tools, and skills.
- **Single web-search owner.** Only market-researcher holds the `web_search` tool (LinkUp primary, Tavily fallback); every other subagent reads `/research/market_brief.md` instead of searching independently.
- **Concurrent delegation.** salary-intelligence and market-tracker are delegated in the same orchestrator step — both are independent of each other, so they execute together instead of sequentially.
- **Skills as progressive disclosure.** salary-benchmarking and roadmap-planning `SKILL.md` methodology guides load on demand, keeping the base system prompt lean.

Backends & persistence

- **CompositeBackend routing.** `/profile`, `/research`, `/analysis`, `/roadmap` resolve to real files on disk (`FilesystemBackend`); `/memories/*` resolves to a `Store` instead.
- **Cross-session memory.** `career_progress.md` persists in an `InMemoryStore` (swap for a Postgres-backed store in production) so a new thread can recall prior findings.
- **Checkpoint.** A `MemorySaver` holds per-thread state — required for the human-in-the-loop interrupt to pause and later resume mid-run.
- **Stateless subagents.** Each task call is a fresh run with no memory of prior subagent turns — instructions must be self-contained.

Validation, HITL & safety

- **Deterministic validators.** `roadmap_validator` and `risk_dashboard_validator` check required sections and 0-10 score ranges in code, not by asking the model to self-report.
- **Human-in-the-loop gate.** `interrupt_on={"sync_to_calendar": True}` pauses before every simulated calendar sync; the run resumes only after an explicit approve/reject decision.
- **Critic pass.** A dedicated critic subagent checks grounding accuracy, skill-methodology compliance, and cross-file consistency before the run is considered complete.
- **Placeholder-key guard.** Unfilled `.env.example` values are explicitly detected and treated as “not configured”, not as a live credential.

Resilience & observability

- **Bounded retries.** The chat model is constructed explicitly with `max_retries=2`, `timeout=60s` — a rate-limit hit fails fast and visibly instead of silently backing off for minutes.
- **Streamed live progress.** The Streamlit UI streams `agent.stream(..., stream_mode="values")` into a status panel instead of a static spinner, surfacing each delegation as it happens.
- **Structured trace.** `build_trace_entries()` parses raw messages into a labeled, sequential trace — planning, delegation, subagent result, final answer — instead of a flat role/content dump.
- **Optional LangSmith tracing.** Setting `LANGSMITH_API_KEY` uploads a full nested trace tree: orchestrator, each task child span, and the subagent's internal LLM + tool calls.

AT A GLANCE

8

specialist subagents

2

operating modes

5

risk axes scored

2x

search providers,
primary + fallback

≤60s

bounded LLM call
timeout

1

HITL gate before any
simulated external
action